

R2P Program Unit 4 - Lesson 2

Quantum Simulation

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IBM Quantum

Outline

1. Core Concepts
 - Postulates of Quantum Mechanics
2. Quantum Simulation
 - Eigenstate Solvers
 - Simulating Dynamics
 - Phase Transitions
3. Classical Approaches
4. Simulating Quantum Systems with QPUs
 - Preparing Quantum Simulation Algorithms
 - Mapping to QPUs
5. Approaches to QPU-based Quantum Simulation
 - VQE
 - SQD
 - Phase Estimation
 - Trotter Approximation
6. Practical Example

Core Concepts

Core Concepts

Postulates of quantum mechanics

The entire formalism of quantum mechanics rests upon five basic postulates, which enable us to understand:

- *How a quantum state is described at any given time t*
- *How to calculate the various physical quantities from this quantum state*
- *How to follow the time evolution of a given quantum state*

The mathematical tools we describe later are what we will use to investigate how quantum systems evolve as well as how to characterize them.

Core Concepts

Postulates of quantum mechanics

Postulate 1 (State of a system):

The state of any physical system, at any time, is specified by a **state vector**: $|\psi(t)\rangle$

in a Hilbert space: $\mathcal{H}$

The system is completely described by this state vector. And any superposition of state vectors is *also* a state vector.

Core Concepts

Postulates of quantum mechanics

Postulate 1 (State of a system):

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Postulate 2 (Observables & Operators):

For every physically measurable quantity: A
(known as an **observable**), there is a corresponding
Hermitian operator: $\hat{A}$
whose eigenvectors are $|\psi_n\rangle$ form a complete basis
and the eigenvalues are a_n .

Core Concepts

Postulates of quantum mechanics

Postulate 1 (State of a system):

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Postulate 2 (Observables & eigenvalues):

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Postulate 3 (Measurements & eigenvalues):

The only possible result of measuring an operator $\hat{A}$ is one of the eigenvalues a_n

Upon measurement, the initial state $|\psi\rangle$ collapses in one of the eigenstates due to the effect of the projection:

$$|\psi_n\rangle\langle\psi_n|\psi(t)\rangle$$

Projection onto eigenstate $|\psi_n\rangle$

Core Concepts

Postulates of quantum mechanics

Postulate 1 (State of a system):

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Postulate 3 (Measurements & eigenvalues):

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Postulate 4 (Probabilistic outcome of measurement):

When making a measurement on a state
the probability of obtaining one of the
nondegenerate eigenvalues a_n is given by:

$$P_n(a_n) = \frac{|\langle \psi_n | \psi \rangle|^2}{\langle \psi | \psi \rangle}$$

Core Concepts

Postulates of quantum mechanics

Postulate 1 (State of a system):

The state of a system is represented by a state vector $|\psi\rangle$ in a Hilbert space. The vector is normalized such that $\langle\psi|\psi\rangle = 1$.

Postulate 2 (Observable quantities):

For any observable quantity A , there exists a Hermitian operator $\hat{A}$ acting on the state vector.

Postulate 3 (Measurements & eigenvalues):

The only possible results of a measurement of A are the eigenvalues a_i of the operator $\hat{A}$.

Upon measurement, the state vector collapses into one of the eigenstates $|a_i\rangle$ corresponding to the measured value a_i .

Postulate 4 (Probabilistic outcome of measurement):

When a measurement of A is performed on a system in state $|\psi\rangle$, the probability of obtaining the value a_i is given by the squared magnitude of the projection of $|\psi\rangle$ onto the eigenstate $|a_i\rangle$.

Postulate 5 (Time evolution):

The time evolution of the state vector is governed by the time-dependent **Schrödinger equation**

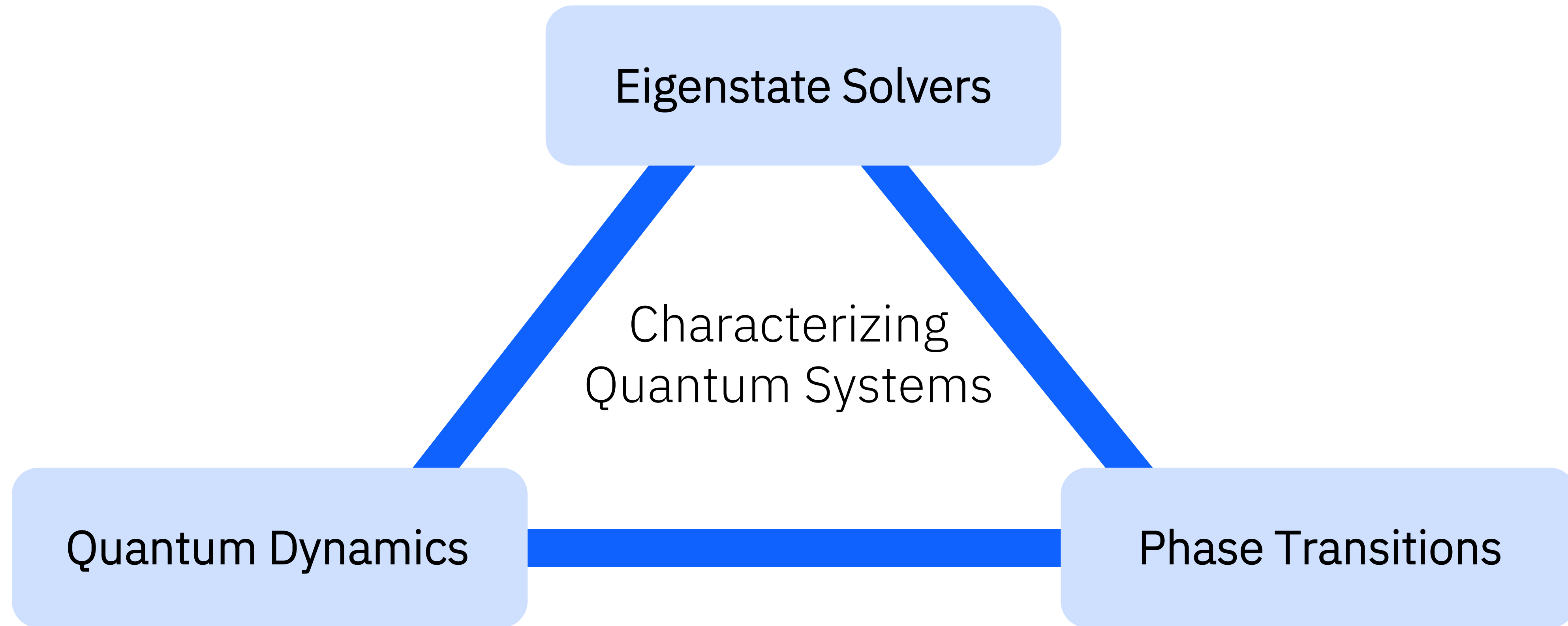
$$i\hbar \frac{d|\psi\rangle}{dt} = H|\psi\rangle$$

Which has the general solution:

$$|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$$

Quantum Simulations

Quantum Simulation



Quantum Simulation

Eigenstate Solvers

Definition: Algorithms to find eigenstates and eigenvalues of a Hamiltonian H .

$$H|\psi\rangle = E|\psi\rangle$$

Why important?

- Ground state energy $\rightarrow$ chemistry, materials.
- Excited states $\rightarrow$ spectroscopy, reaction dynamics.

Challenge: As the system grows, the Hilbert space exponentially grows.

Straightforward approach but quickly becomes intractable as system size grows.

Quantum Simulation

Eigenstate Solvers

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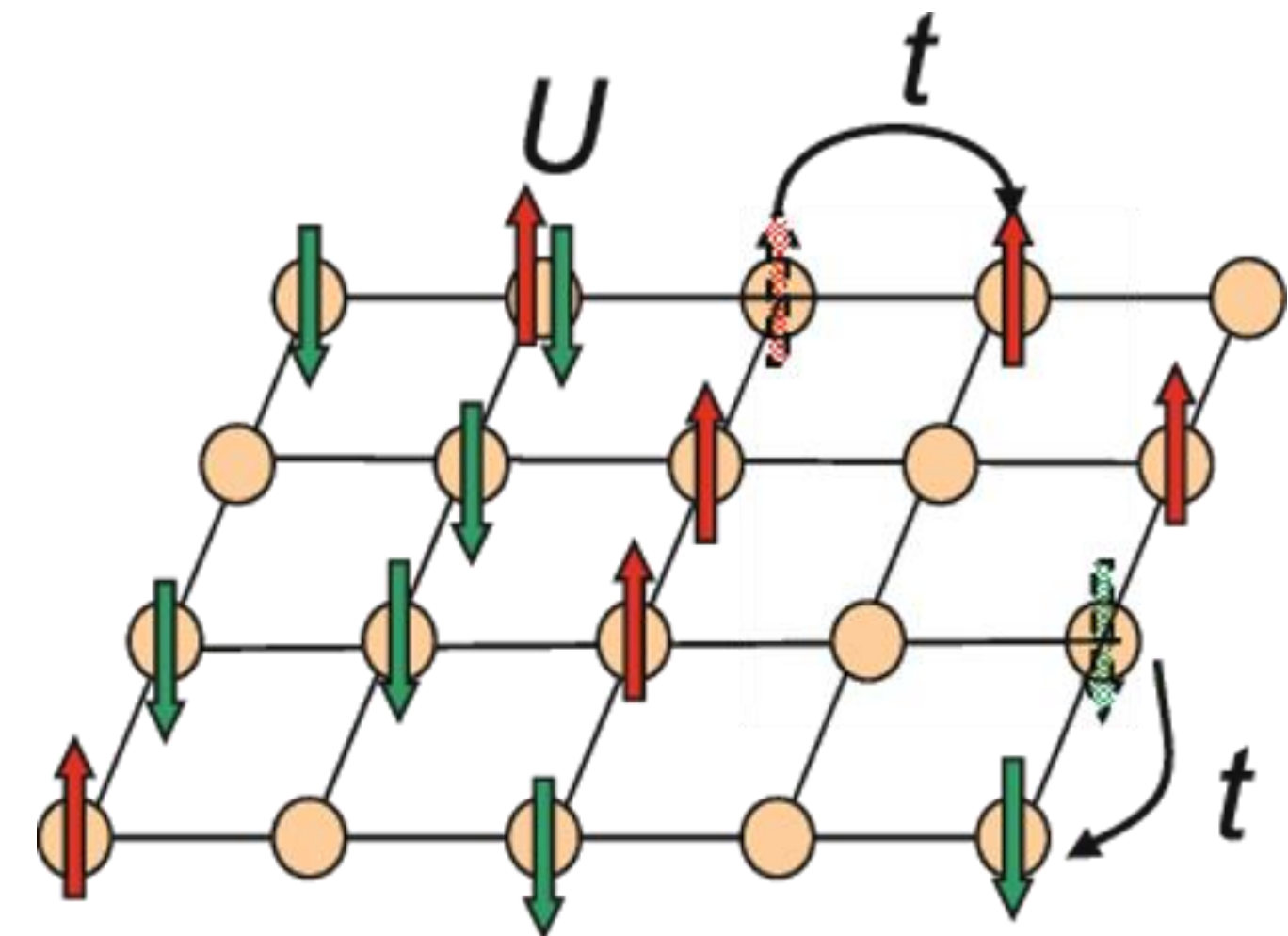
Straightforward approach but quickly becomes intractable as system size grows.

Classical Methods:

- Exact diagonalization (only for small systems)
- Approximate methods:
 - Tensor networks (DMRG)
 - Iterative solvers (LOBPCG, Davidson)

Quantum Methods:

- VQE: Hybrid, variational (suitable for pre-fault tolerant hardware)
- QPE: Precise (needs fault-tolerant hardware)



Susumu Yamada, et. al. *High Performance LOBPCG Method for Solving Multiple Eigenvalues of Hubbard Model: Efficiency of Communication Avoiding Neumann Expansion*

Quantum Simulation

Quantum Dynamics

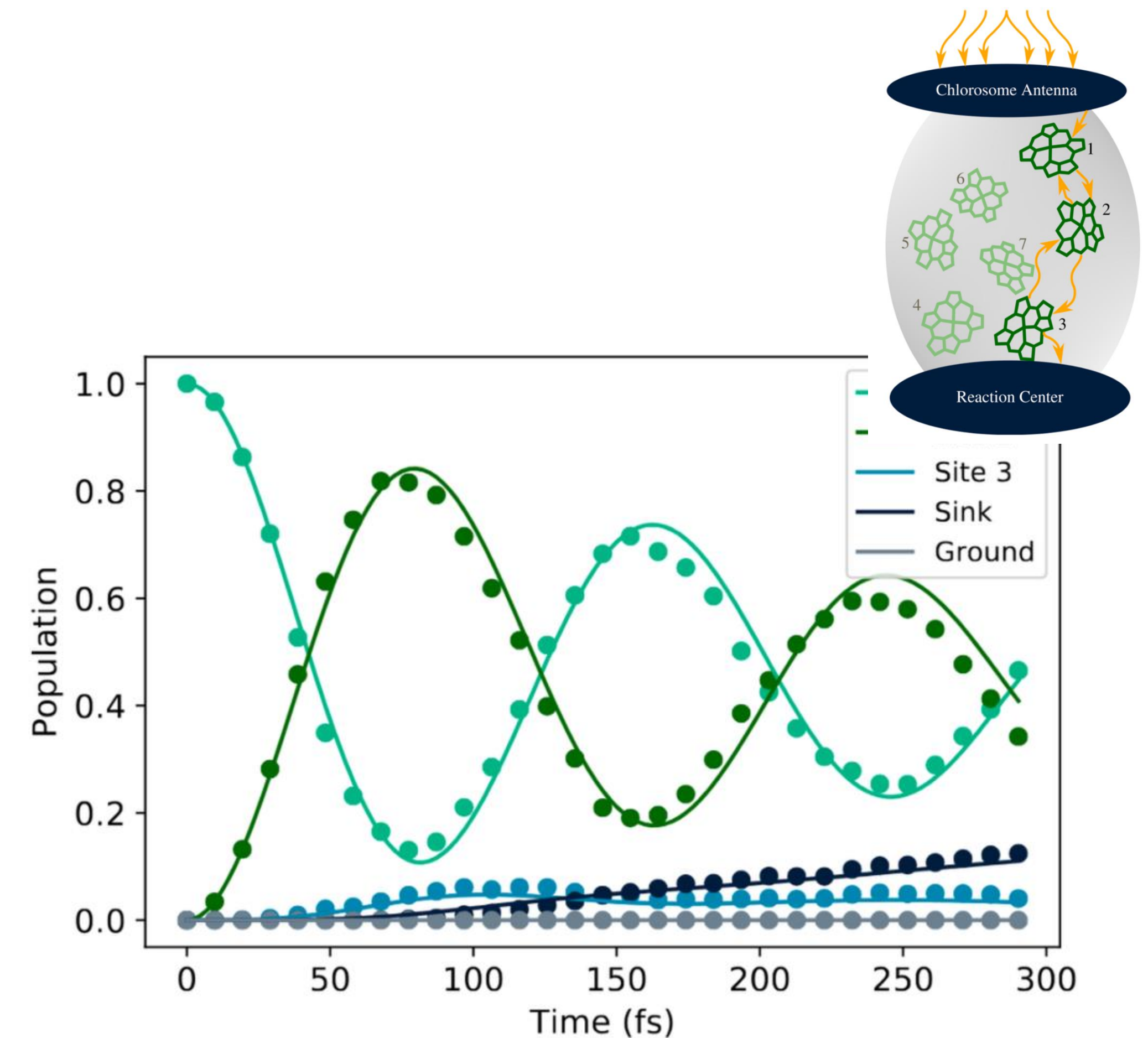
Definition: Study of the time evolution of a quantum system described by a Hamiltonian H where the dynamic is obtained by solving the time-dependent [Schrödinger equation](#):

$$i\hbar \frac{d|\psi\rangle}{dt} = H|\psi\rangle \rightarrow |\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle$$

Where can this be used?

- Real-time dynamics in chemistry and materials
- Non-equilibrium processes
- Transport phenomena

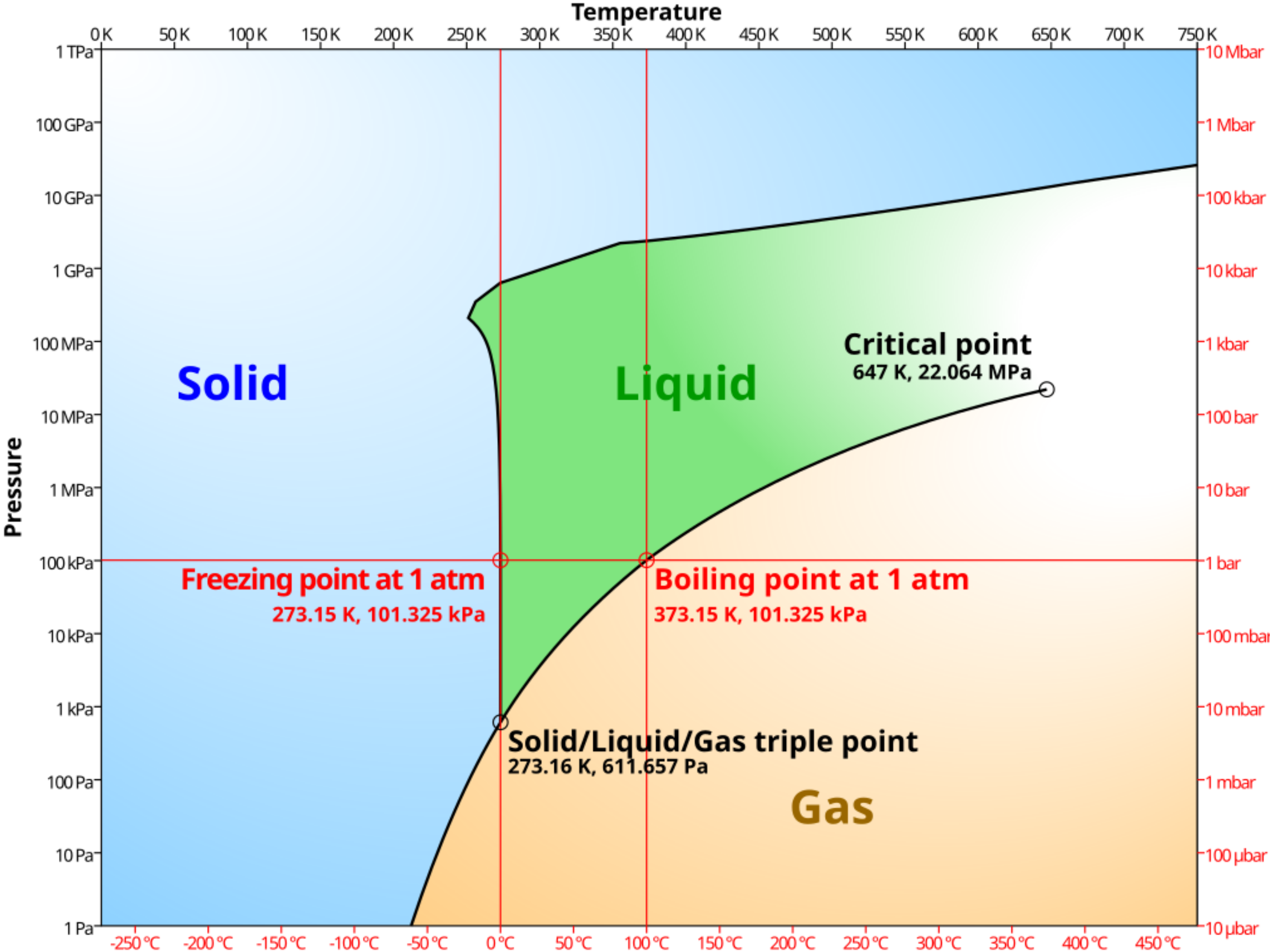
Challenges: Exponential complexity, entanglement growth (for classical approaches).



Quantum 6, 726 (2022)

Quantum Simulation

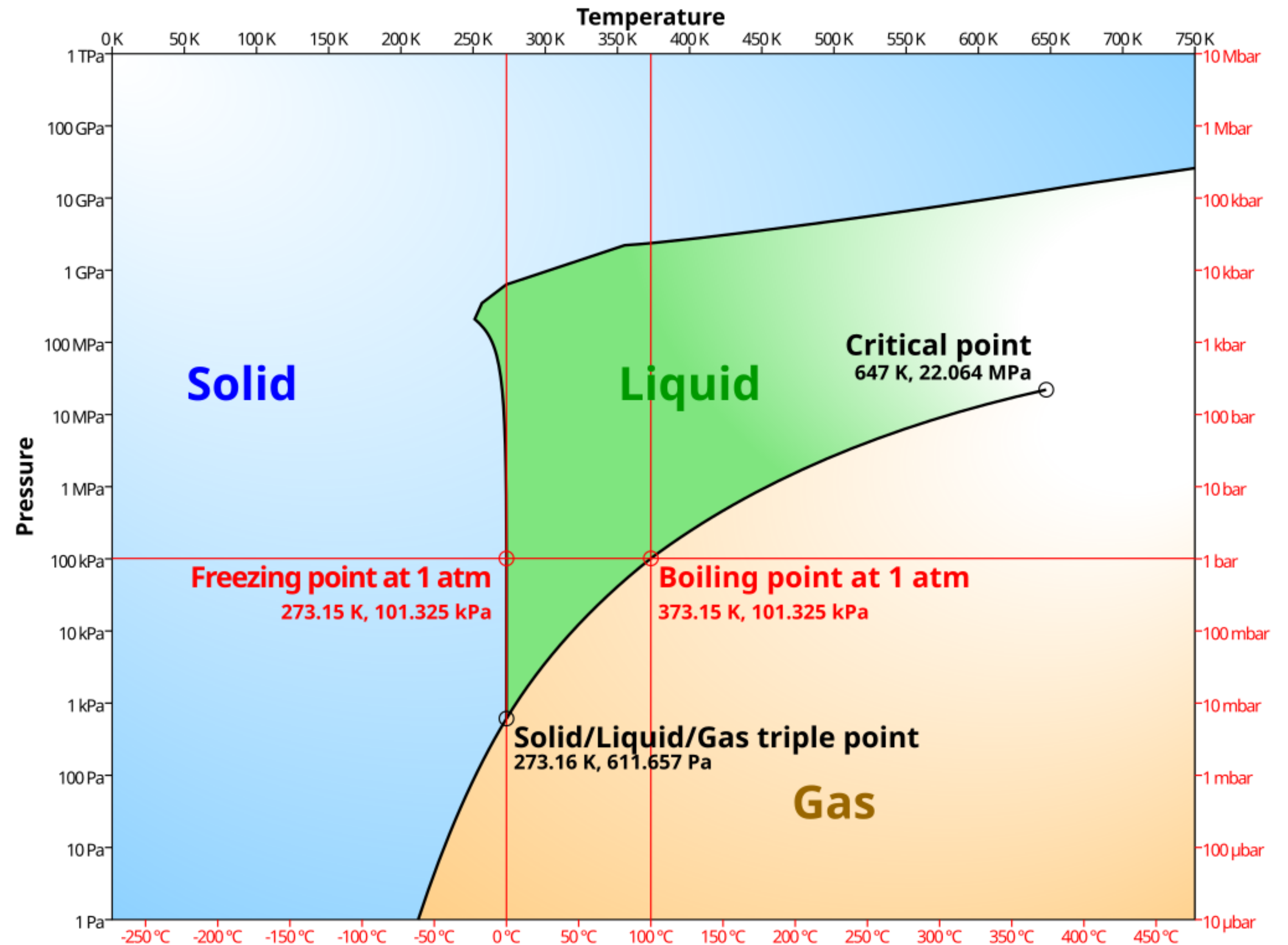
Phase Transitions



Quantum Simulation

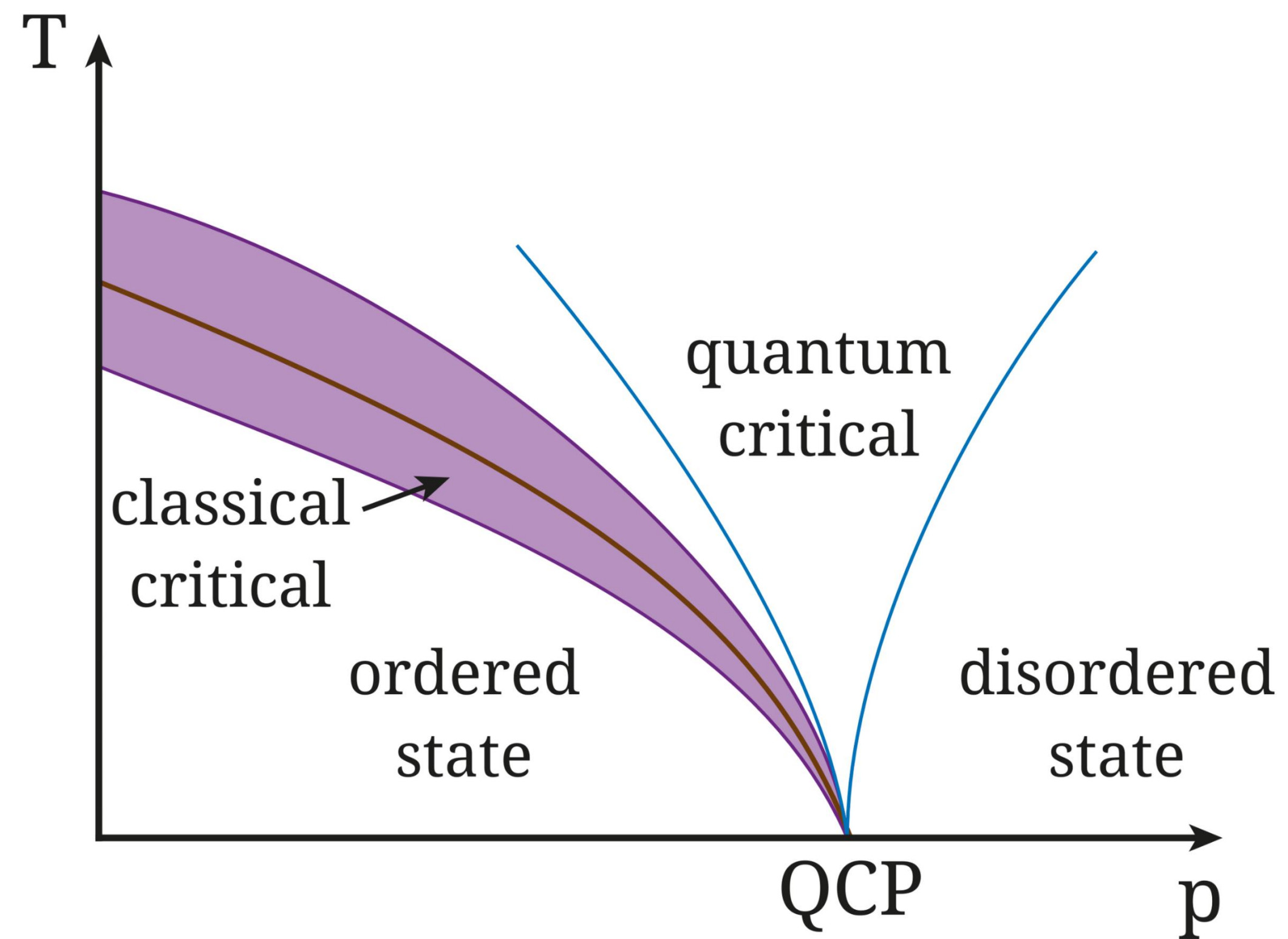
Phase Transitions

Unlike classical phase transitions, *quantum phase* transitions describe abrupt changes in the ground state of a system and are typically induced by varying a magnetic field, temperature, or some other *order parameter*.

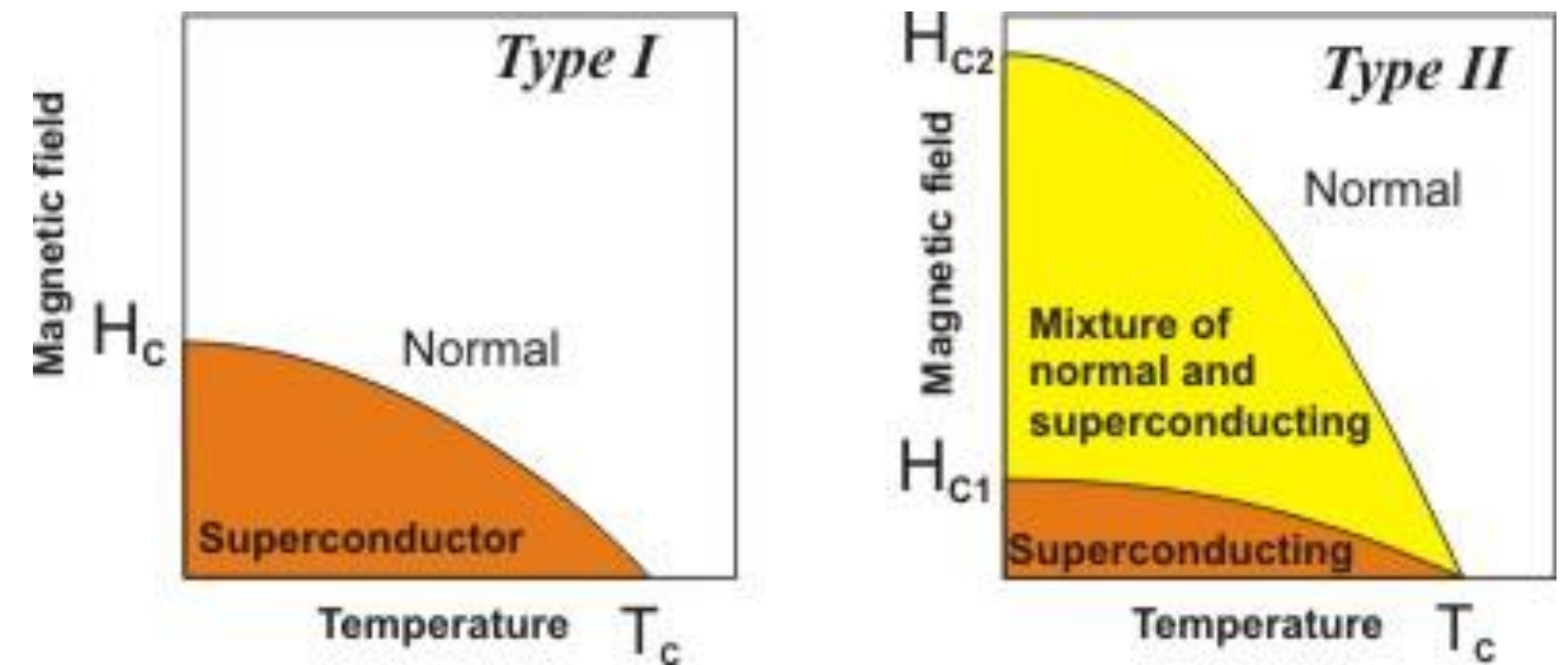


Quantum Simulation

Phase Transitions



Phase transition from conductor to superconductor



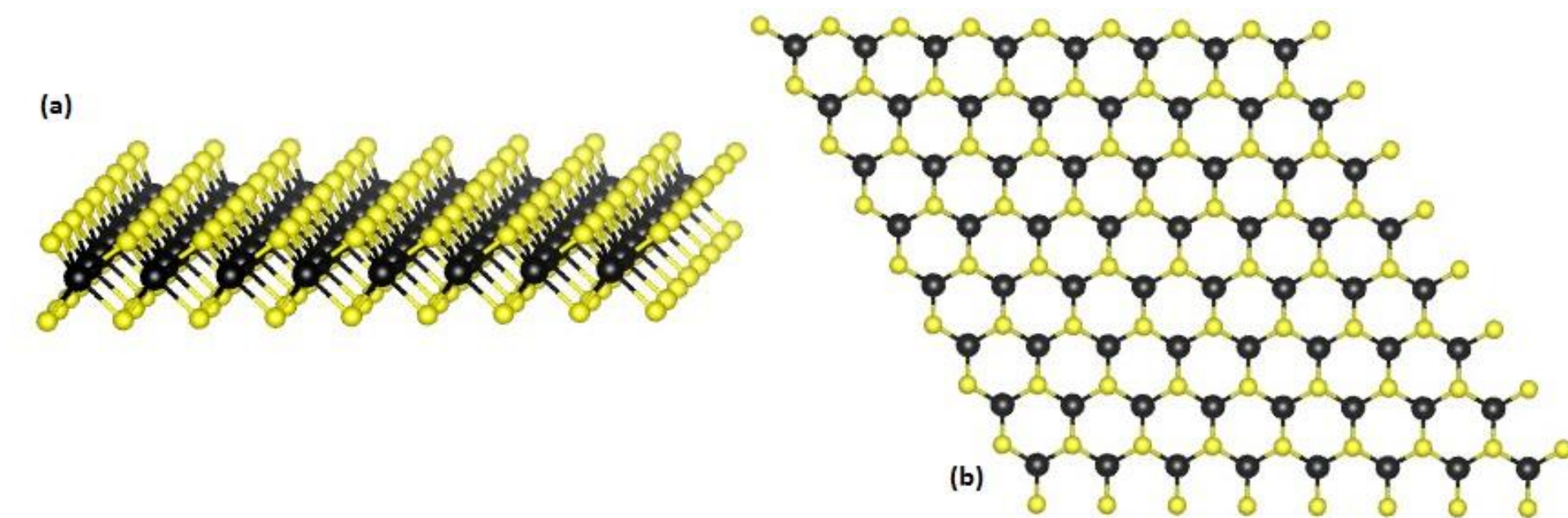
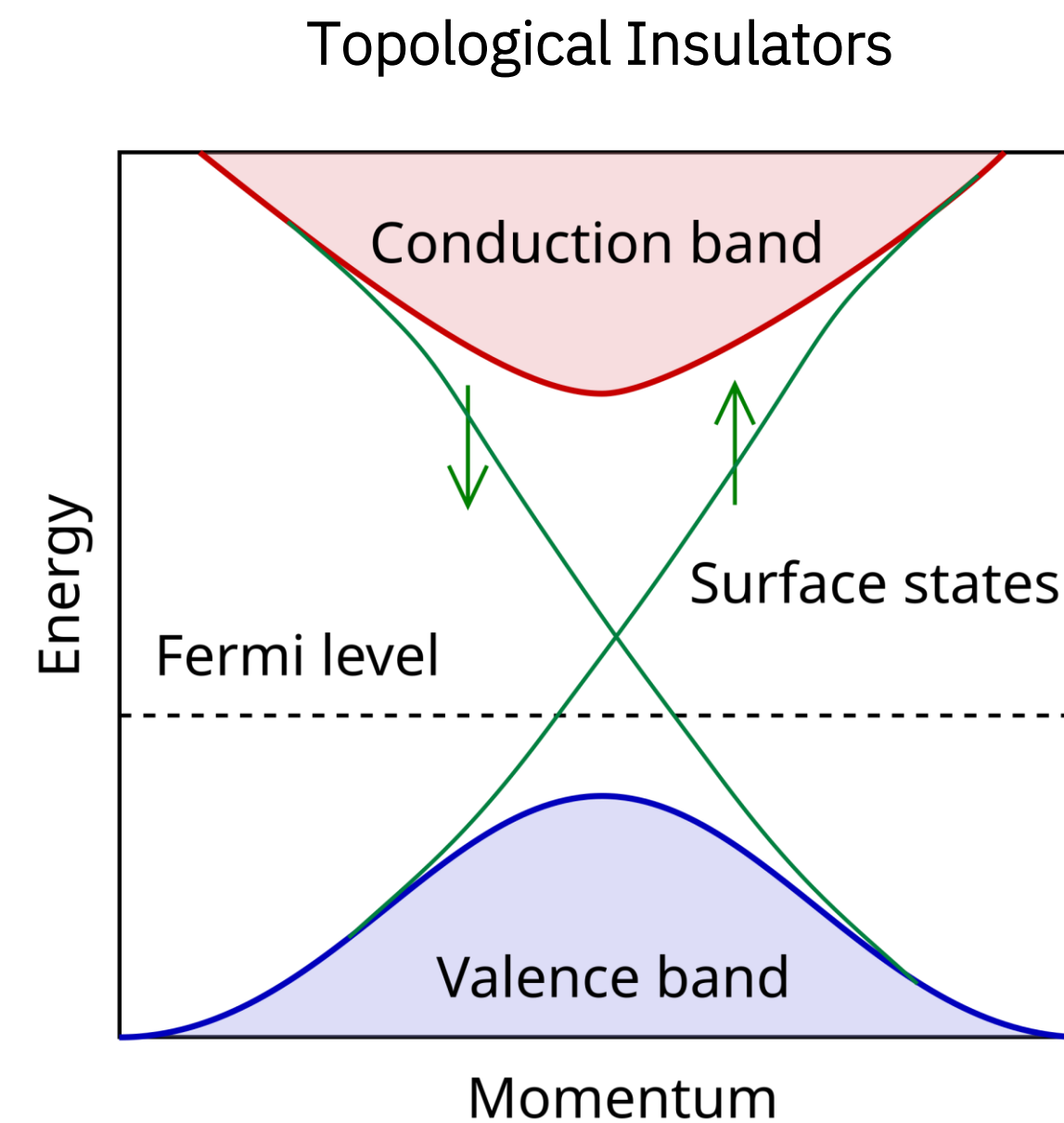
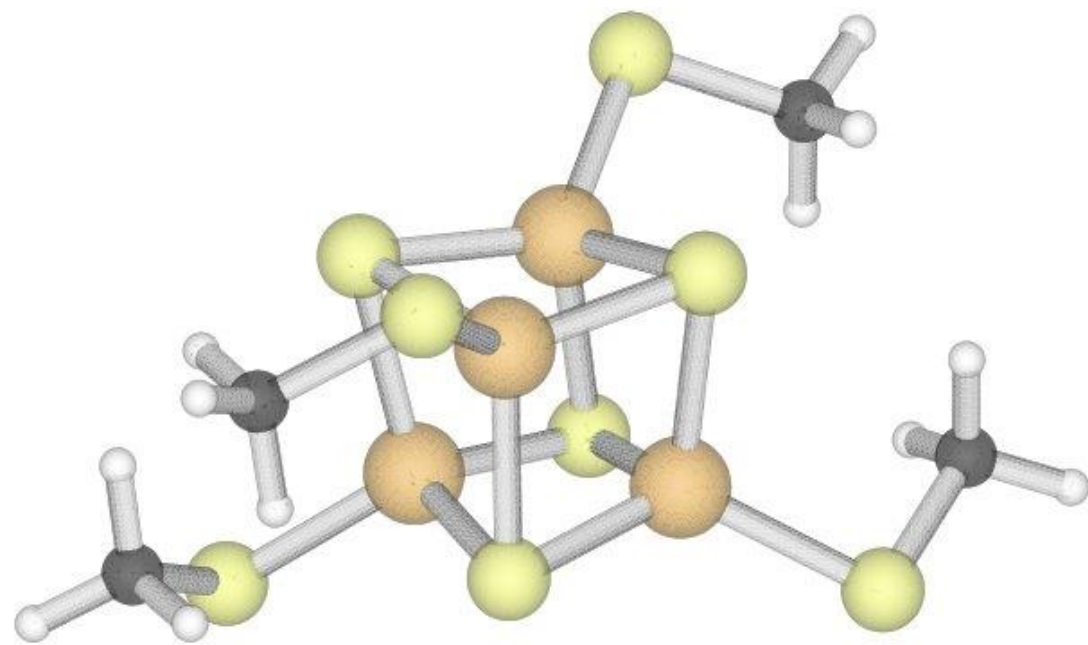
Classical Approaches

Classical Approaches to Quantum Simulation

Classical approaches to Quantum Simulation require fundamental trade-offs because they approximate the system without the knowledge of a full wavefunction.

Popular methods include:

- Tensor Networks (TN)
- Density functional theory (DFT)



Classical Approaches to Quantum Simulation

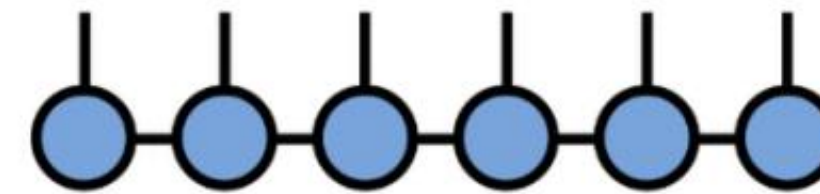
Tensor networks methods

Idea: Compress the wavefunction using low-rank factorization, which can then be then used as Ansatz in variational algorithms.

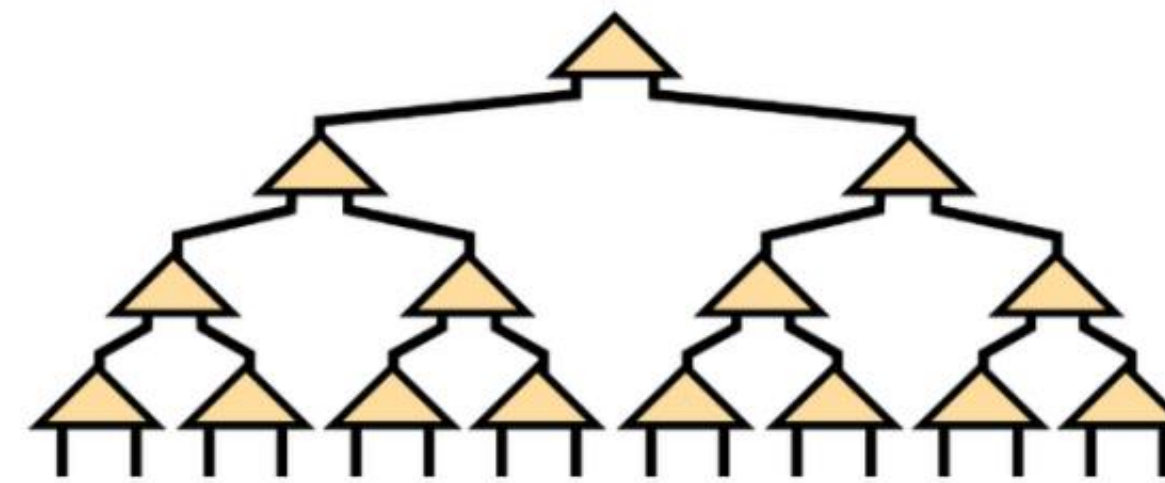
Strengths: Great for strongly correlated systems in 1D and some 2D cases.

Limitations: Struggles with high entanglement and large dimensions

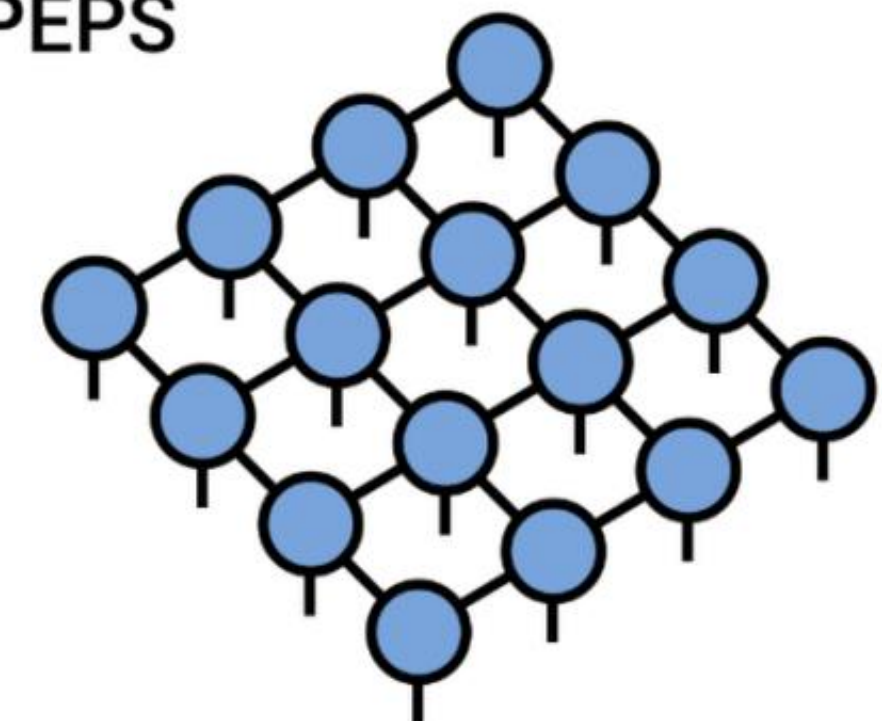
**Matrix Product State /
Tensor Train**



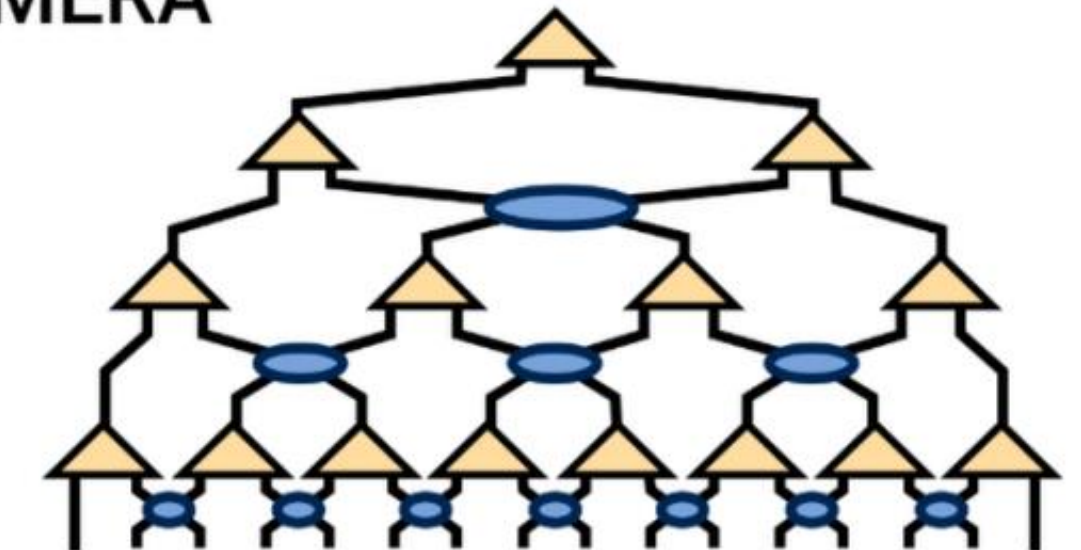
**Tree Tensor Network /
Hierarchical Tucker**



PEPS



MERA



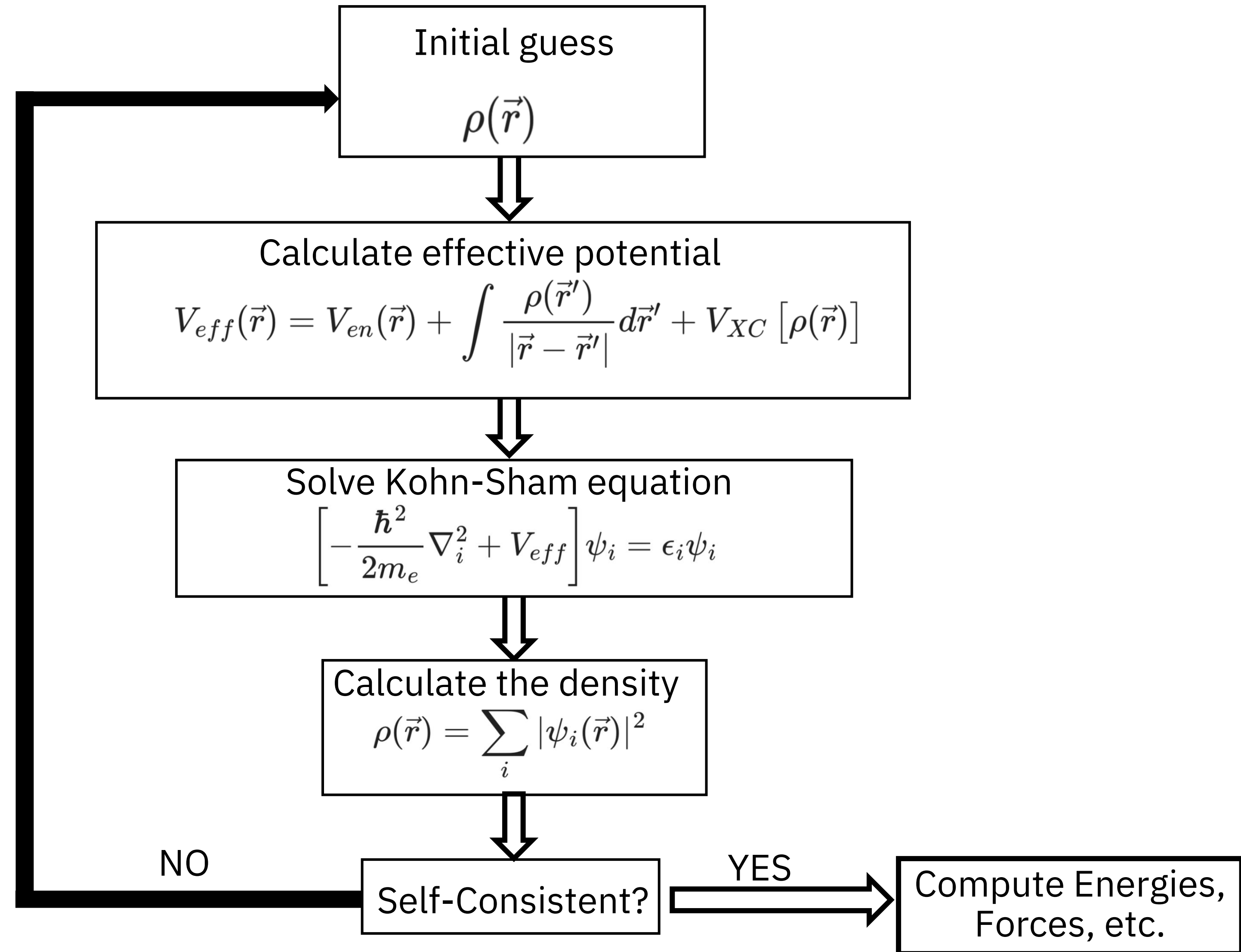
Classical Approaches to Quantum Simulation

Density Functional Theory (DFT)

Idea: Based on the mean-field approach, it replaces the many-body wavefunction $|\psi(\vec{r})\rangle$ with an electron density $\rho(\vec{r})$

Strengths: Scales well with system size and, therefore, is widely used in chemistry and material simulations.

Limitations: Struggles with strongly correlated systems where electron interactions are significant and relies on approximate exchange-correlation functionals.



Simulating Quantum Systems with QPUs

Anatomy of a Quantum Simulation Algorithm

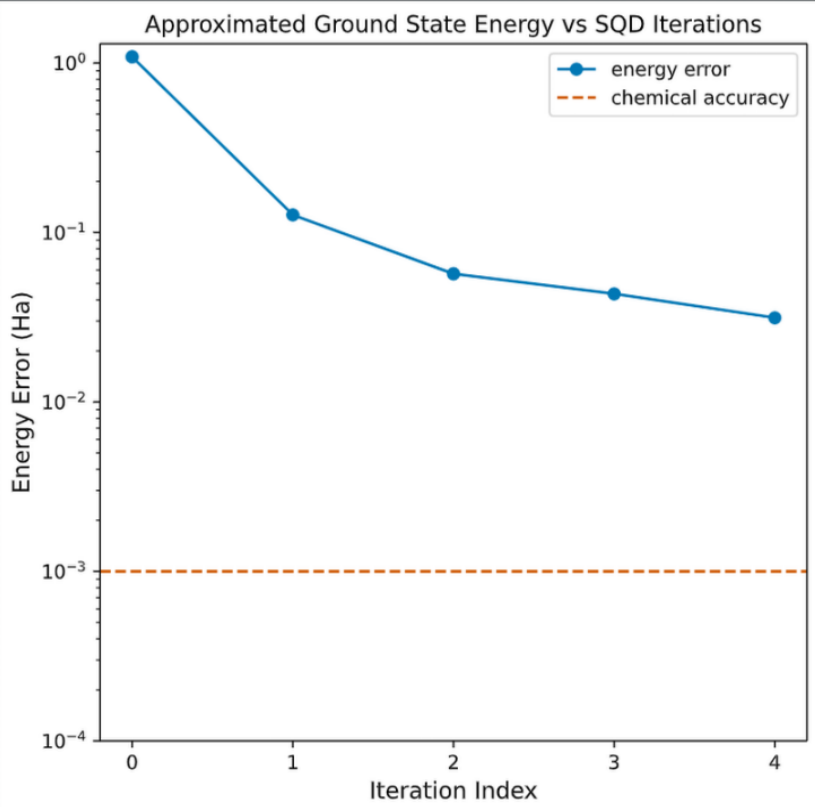
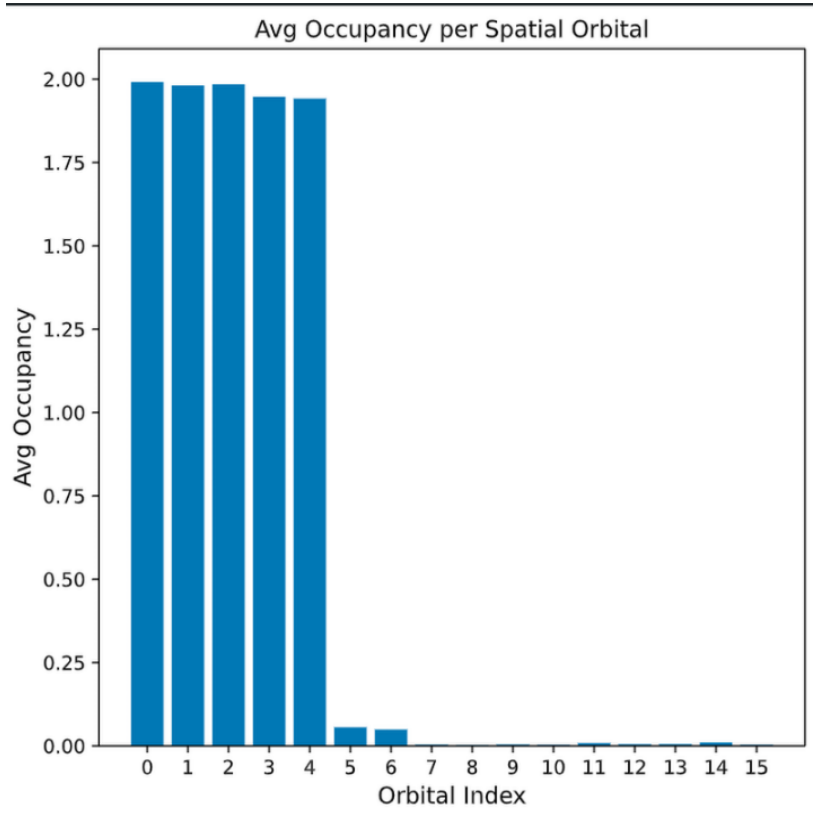
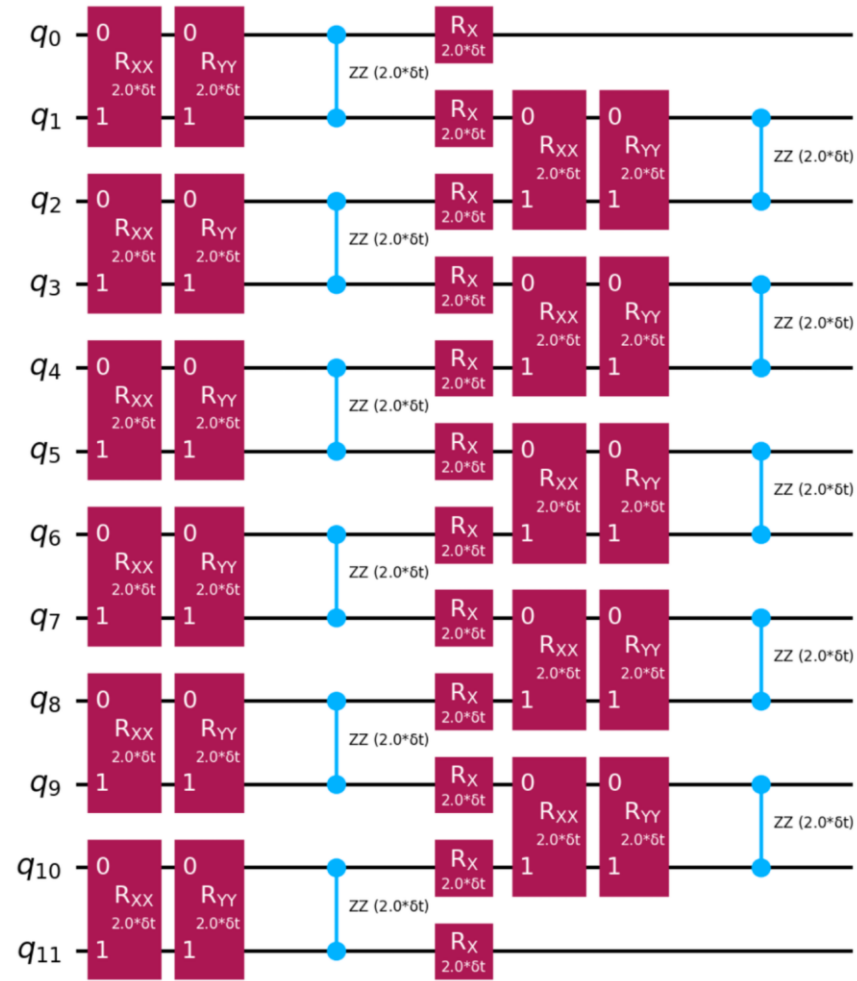
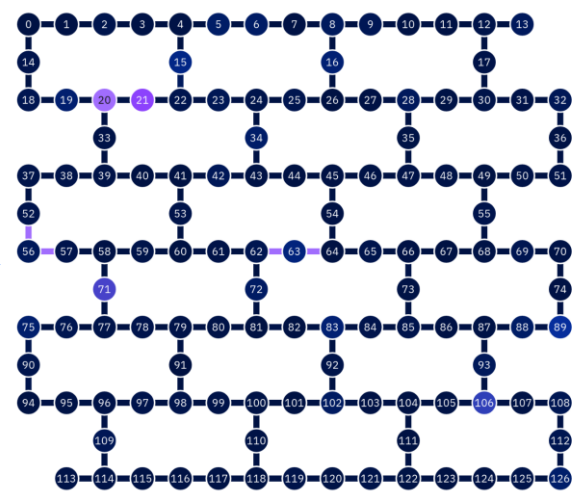
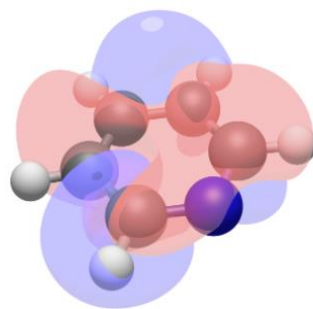
Choose a quantum system to simulate/characterize.
Determine attributes such as the Hamiltonian and any important observables to measure.

Map the Hamiltonian, and the chosen set of basis vectors to a quantum computing environment. This includes rewriting the Hamiltonian in terms of Pauli operators, as well as mapping the states to the computational basis.

Prepare the quantum circuits which will be used for the simulation. This might include choosing an ansatz or creating a time evolution circuit.

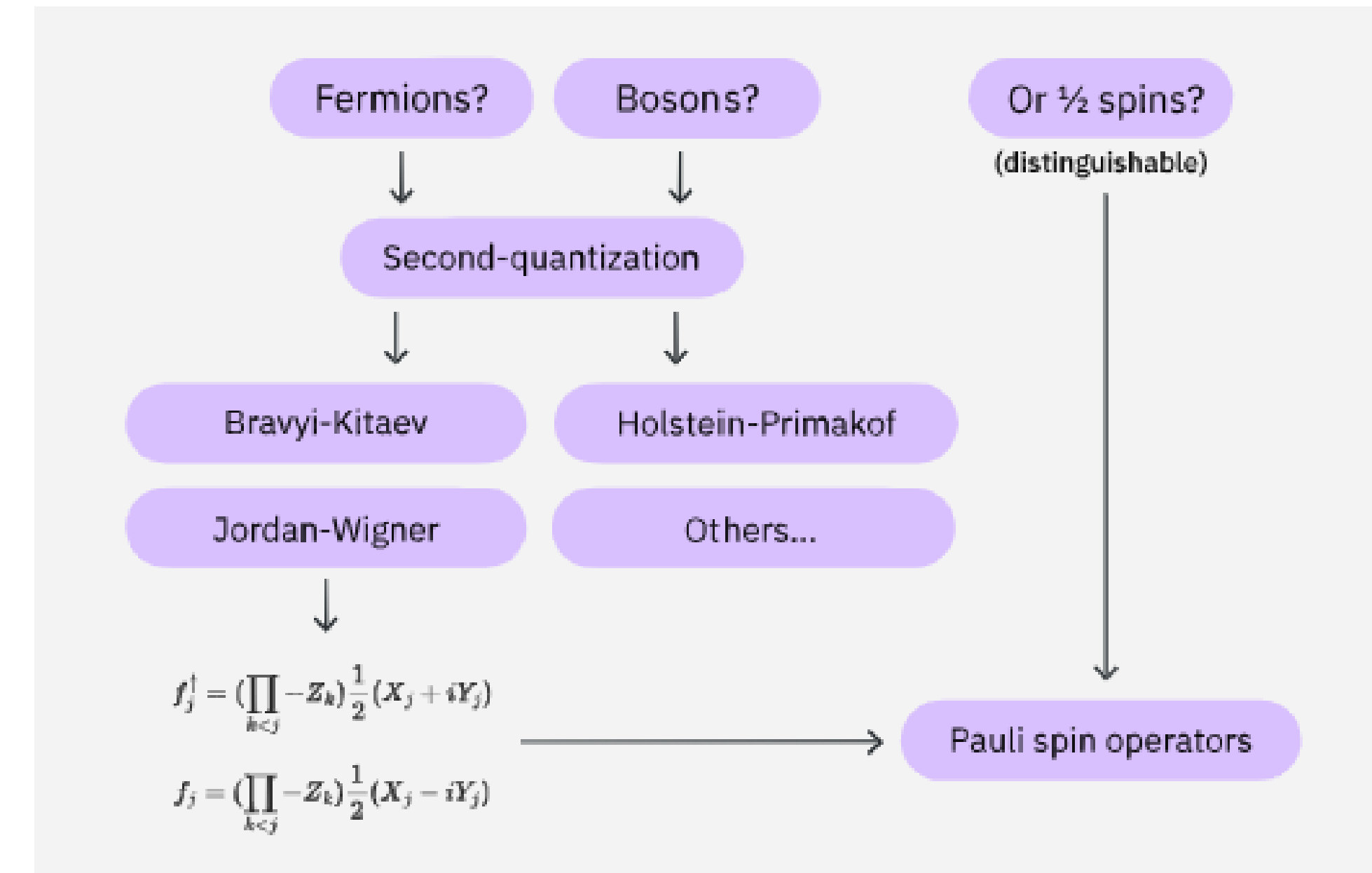
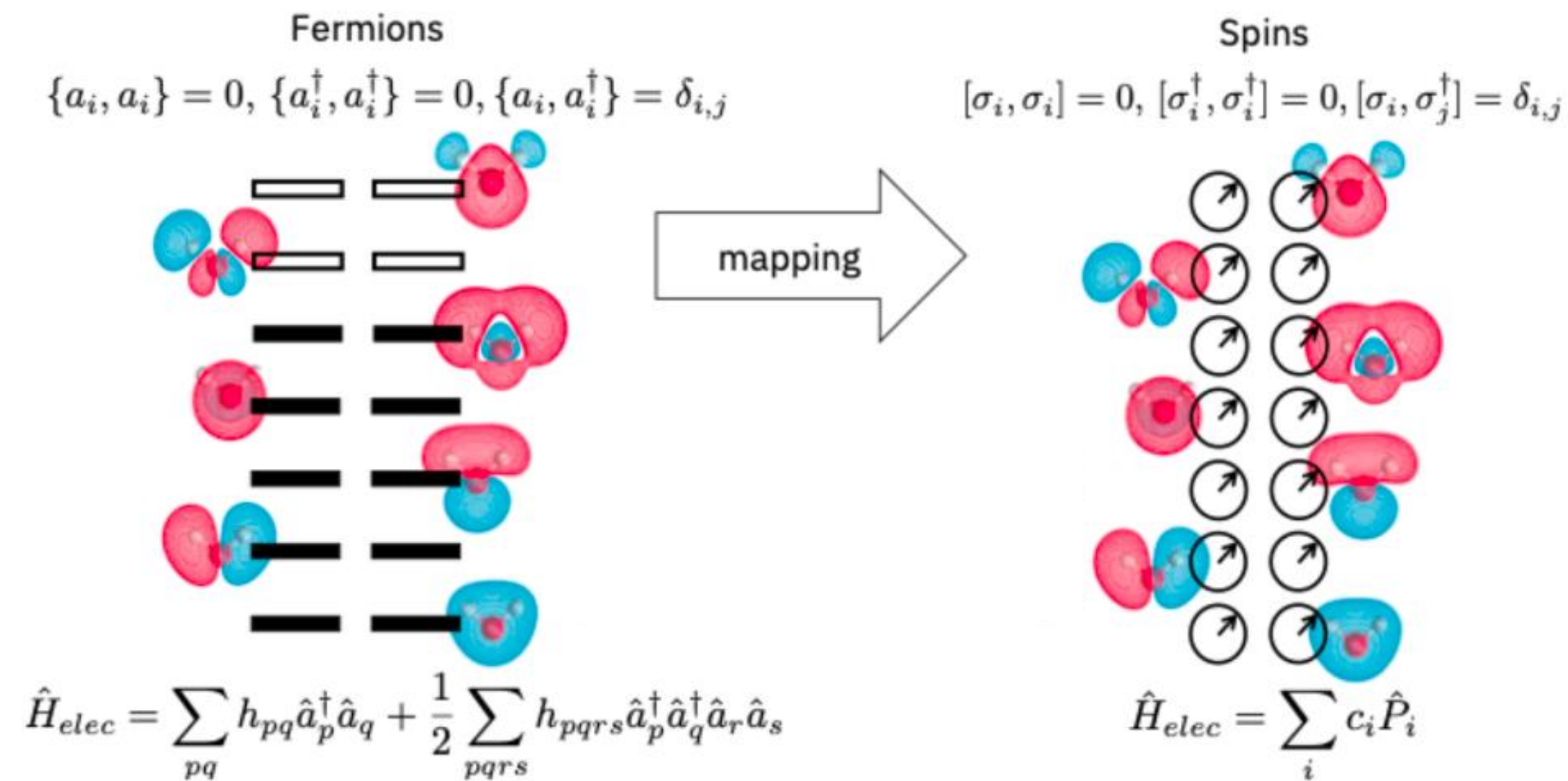
Once the experiments have completed, run any necessary post-processing for either error mitigation or analysis of raw data.

$$H = \sum_i H_i$$



Mapping a problem to QPUs

Hamiltonian encoding



Jordan-Wigner transformation:

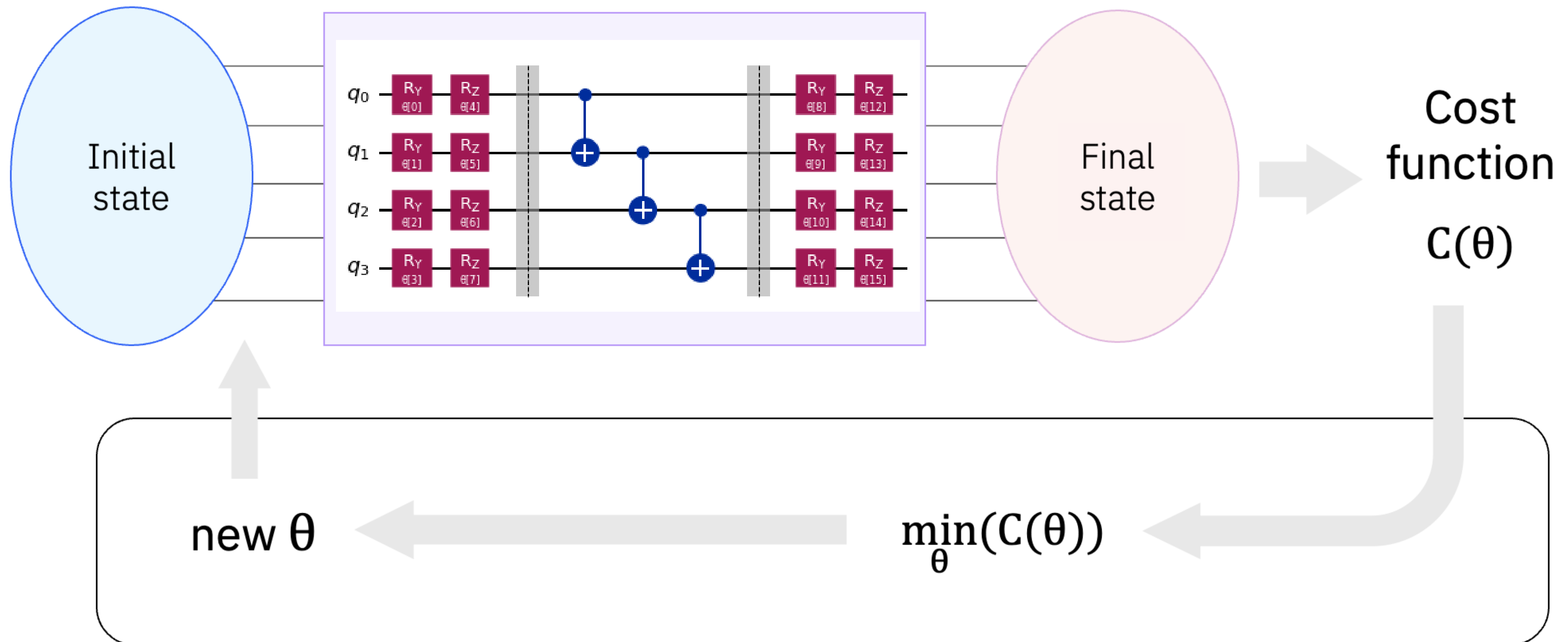
$$\hat{a}_j^\dagger = \prod_{k<j} (-\hat{\sigma}_k^Z) \left(\frac{\hat{\sigma}_k^X - i\hat{\sigma}_k^Y}{2} \right)$$

$$\hat{a}_j = \prod_{k<j} (-\hat{\sigma}_k^Z) \left(\frac{\hat{\sigma}_k^X + i\hat{\sigma}_k^Y}{2} \right)$$

Approaches to QPU-based Quantum Simulations

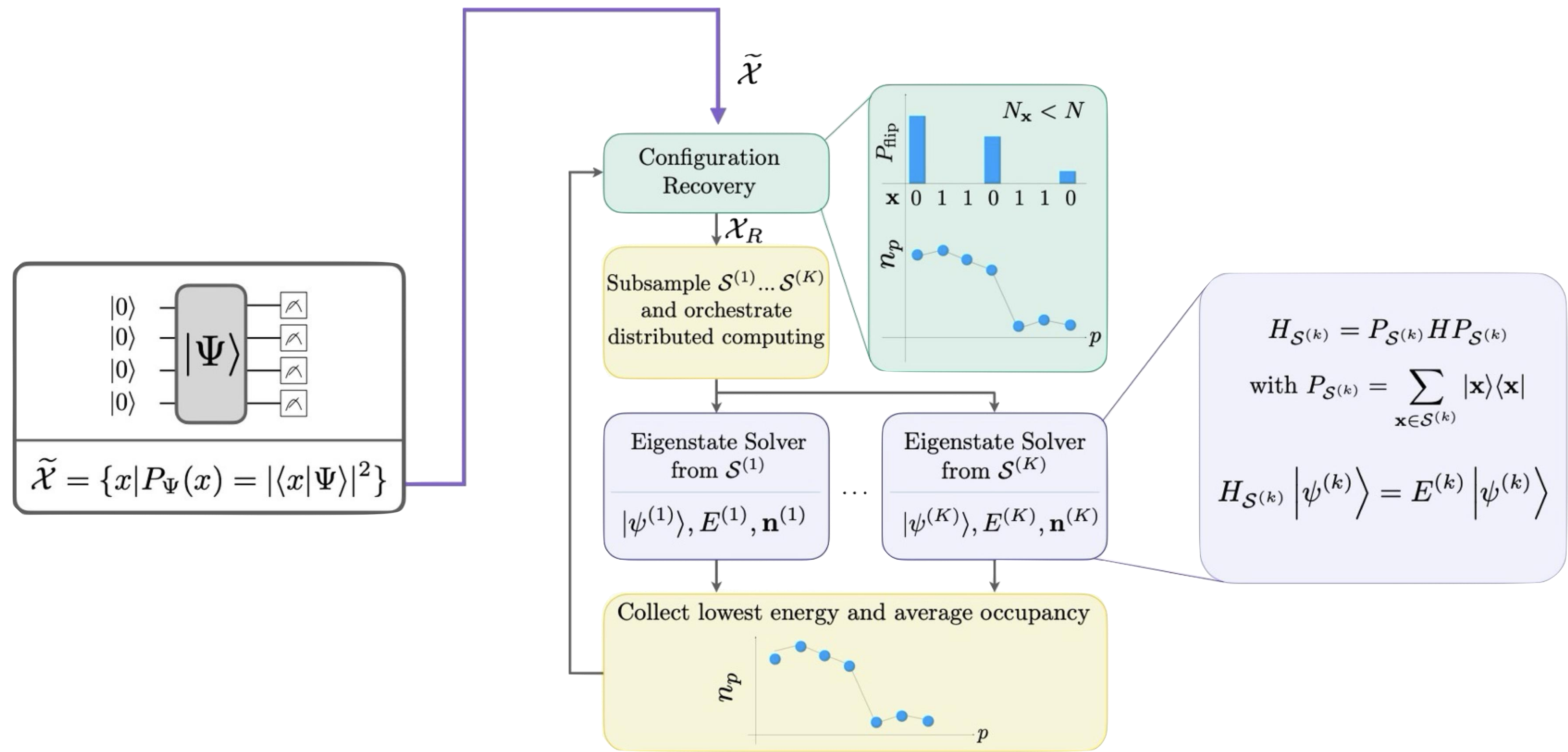
Simulating Quantum Systems

Variational Quantum Eigensolver (VQE)



Simulating Quantum Systems

Sample-based Quantum Diagonalization (SQD)



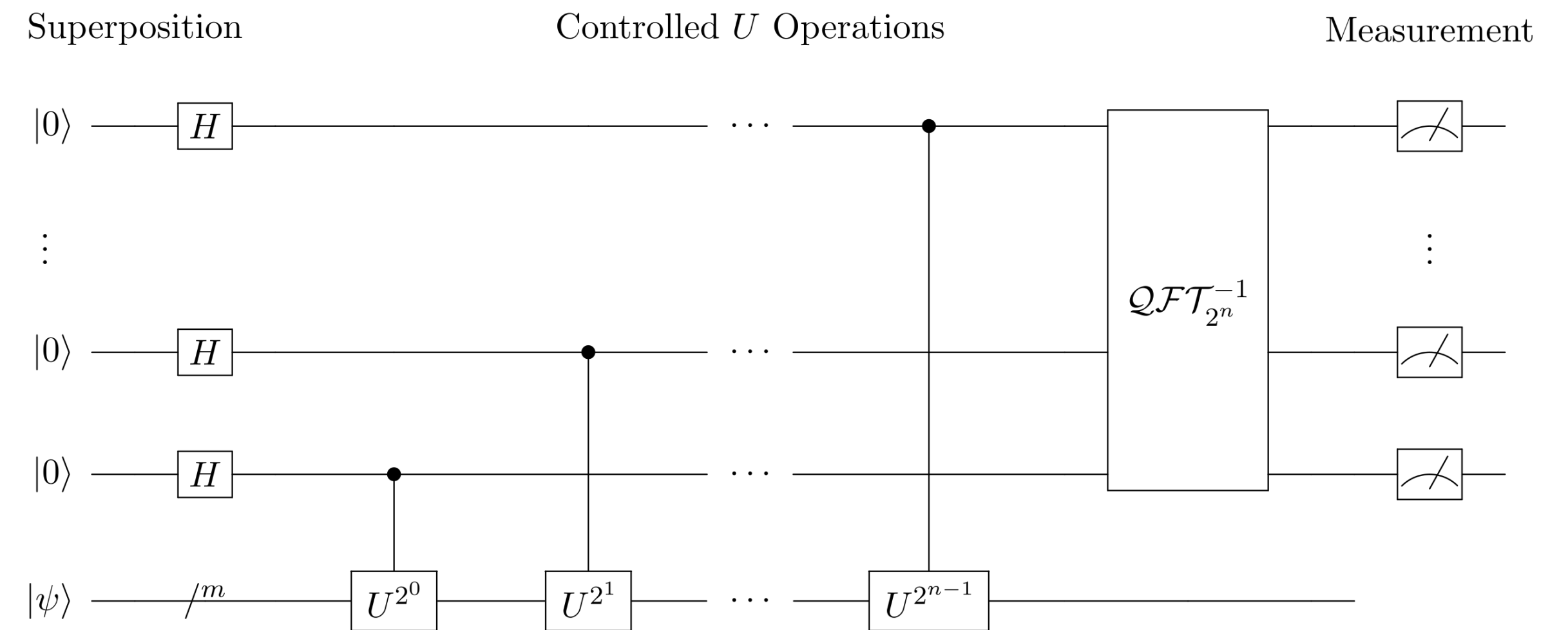
Simulating Quantum Systems

Quantum Phase Estimation

Input: A unitary quantum circuit for a n -qubit operation U and a n -qubit quantum state $|\psi\rangle$

Promise: $|\psi\rangle$ is an eigenvector of U

Output: phase $\theta \in [0,1]$ satisfying:
$$U(t)|\psi\rangle = e^{-2\pi i\theta}|\psi\rangle$$



Several different implementations of QPE: standard QPE, iterative QPE with phase kickback, etc.

Simulating Quantum Systems

(Approximate) Time Evolution

Goal is to implement:

$$U(t) = e^{-iHt}$$

Where: $H = \sum_i H_i$

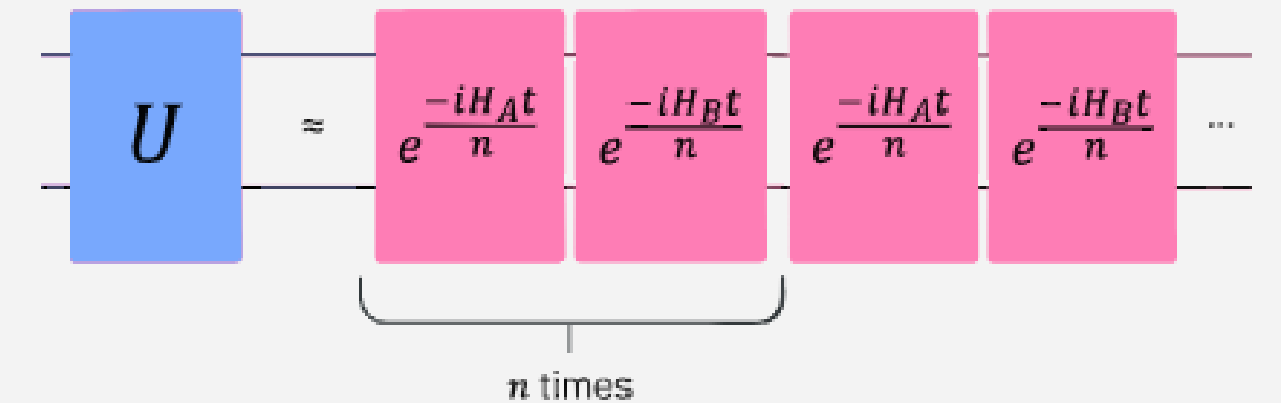
However, if not all terms
in the Hamiltonian commute:

$$e^{-iHt} \neq \prod_i e^{-iH_i t}$$

Lie Product formula:

$$\lim_{n \rightarrow \infty} \left(e^{iAt/n} e^{iBt/n} \right)^n = e^{i(A+B)t}$$

State Evolution - Trotterization



Lie-Trotter (first order) approximation:

$$e^{i(A+B)\Delta t} = e^{iA\Delta t} e^{iB\Delta t} + \mathcal{O}(\Delta t^2)$$

Trotter-Suzuki (second order) approximation:

$$e^{i(A+B)\Delta t} = e^{iA\Delta t/2} e^{iB\Delta t} e^{iA\Delta t/2} + \mathcal{O}(\Delta t^3)$$

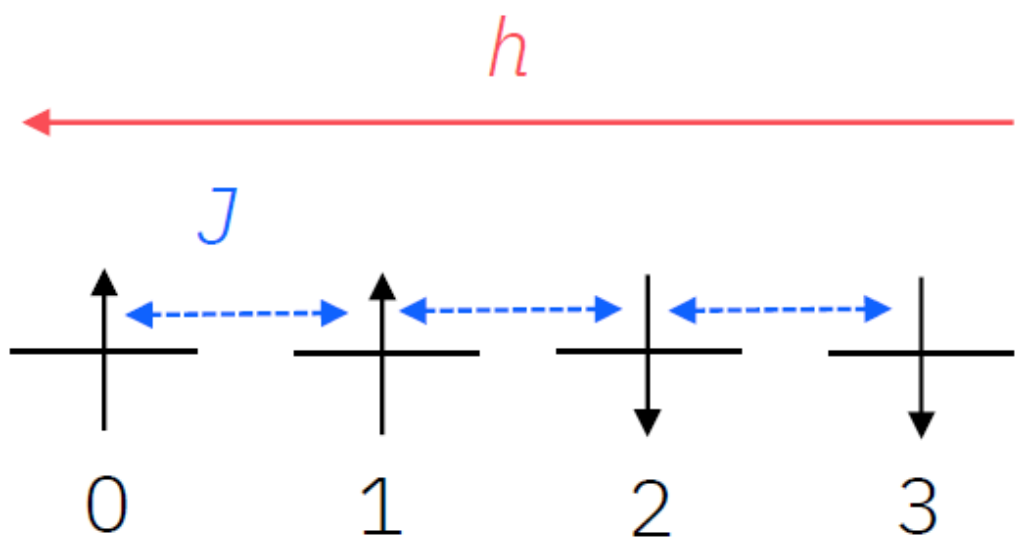
Practical Example

Practical Example: Trotterization (first-order)

Time evolution of 1D transverse-field Ising model

N : Number of qubits

$$H = - \sum_{\langle i,j \rangle}^{N-1} J \sigma_{Z_i} \sigma_{Z_j} - \sum_i^N h_i \sigma_{X_i}$$



Quantum state	Spin representation
$ 0000\rangle$	$\uparrow\uparrow\uparrow\uparrow$
$ 1000\rangle$	$\downarrow\uparrow\uparrow\uparrow$
$\dots$	$\dots$
$ 1111\rangle$	$\downarrow\downarrow\downarrow\downarrow$

$$e^{-i\hat{H}\Delta t} = e^{-i\Delta t(-\sum_{i,j}^N J \sigma_{Z_i} \sigma_{Z_j} - \sum_i^N h_i \sigma_{X_i})} \approx e^{-i\Delta t(-\sum_{i,j}^N J \sigma_{Z_i} \sigma_{Z_j})} e^{-i\Delta t(-\sum_i^N h_i \sigma_{X_i})}$$

$R_{ZZ}(-2J\Delta t)$

$R_X(-2h\Delta t)$

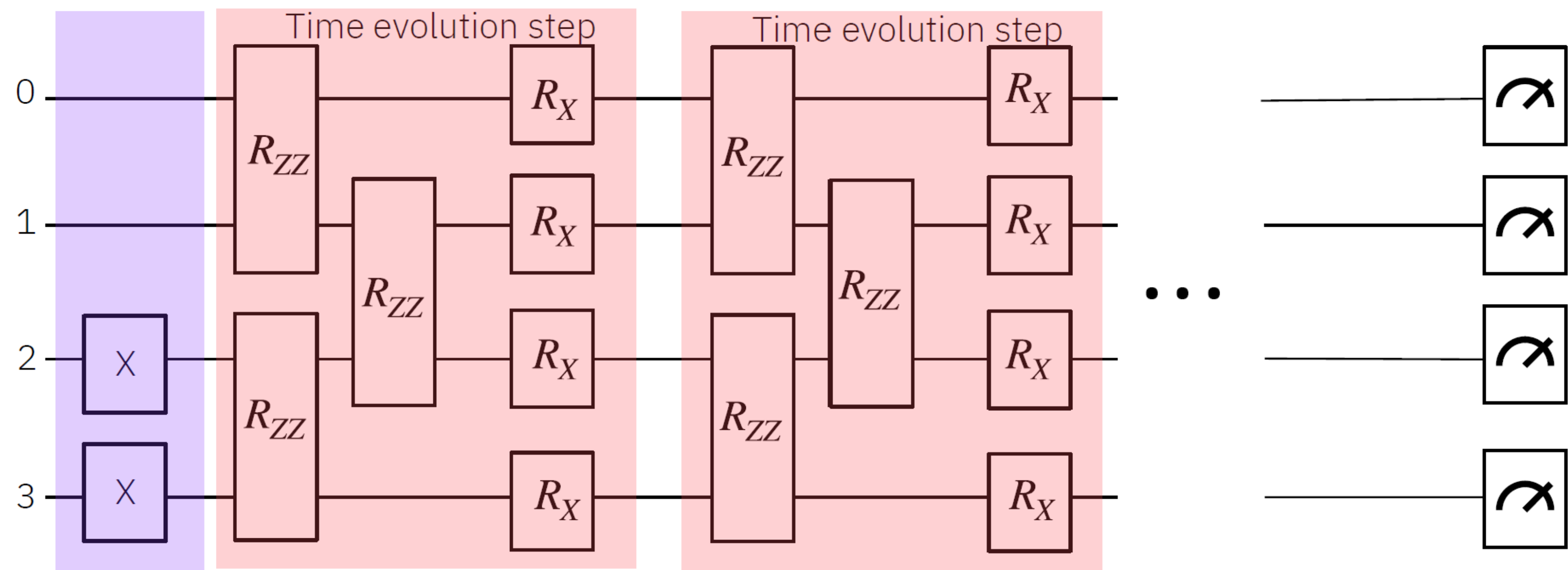
$$R_{ZZ}(\theta) = e^{-i\frac{\theta}{2}\sigma_Z\sigma_Z} = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 & 0 & 0 \\ 0 & e^{i\frac{\theta}{2}} & 0 & 0 \\ 0 & 0 & e^{i\frac{\theta}{2}} & 0 \\ 0 & 0 & 0 & e^{-i\frac{\theta}{2}} \end{pmatrix}$$

$$R_X(\theta) = e^{-i\frac{\theta}{2}\sigma_X} = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

Practical Example: Trotterization (first-order)

Time evolution of 1D transverse-field Ising model

Bit string $|1100\rangle$



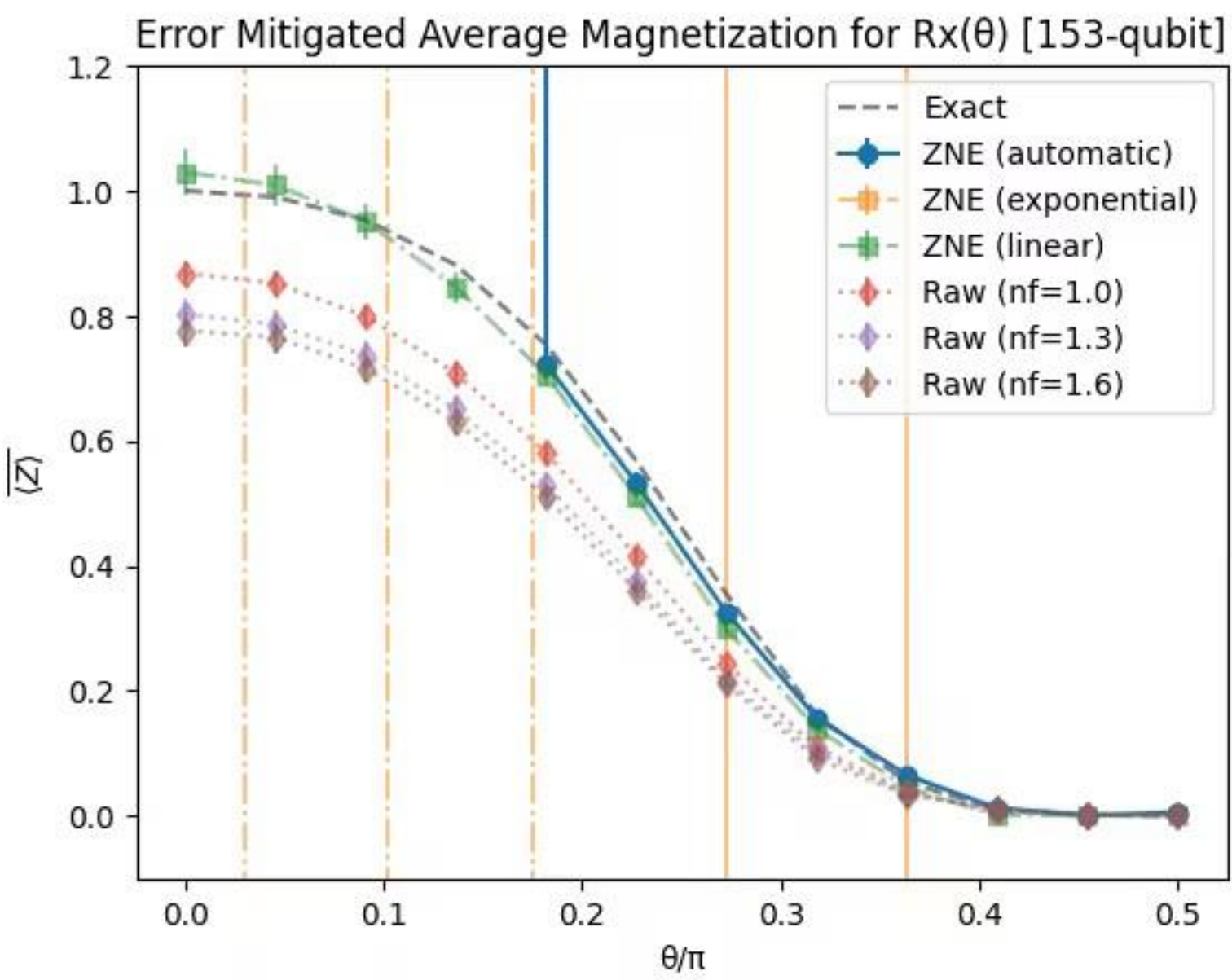
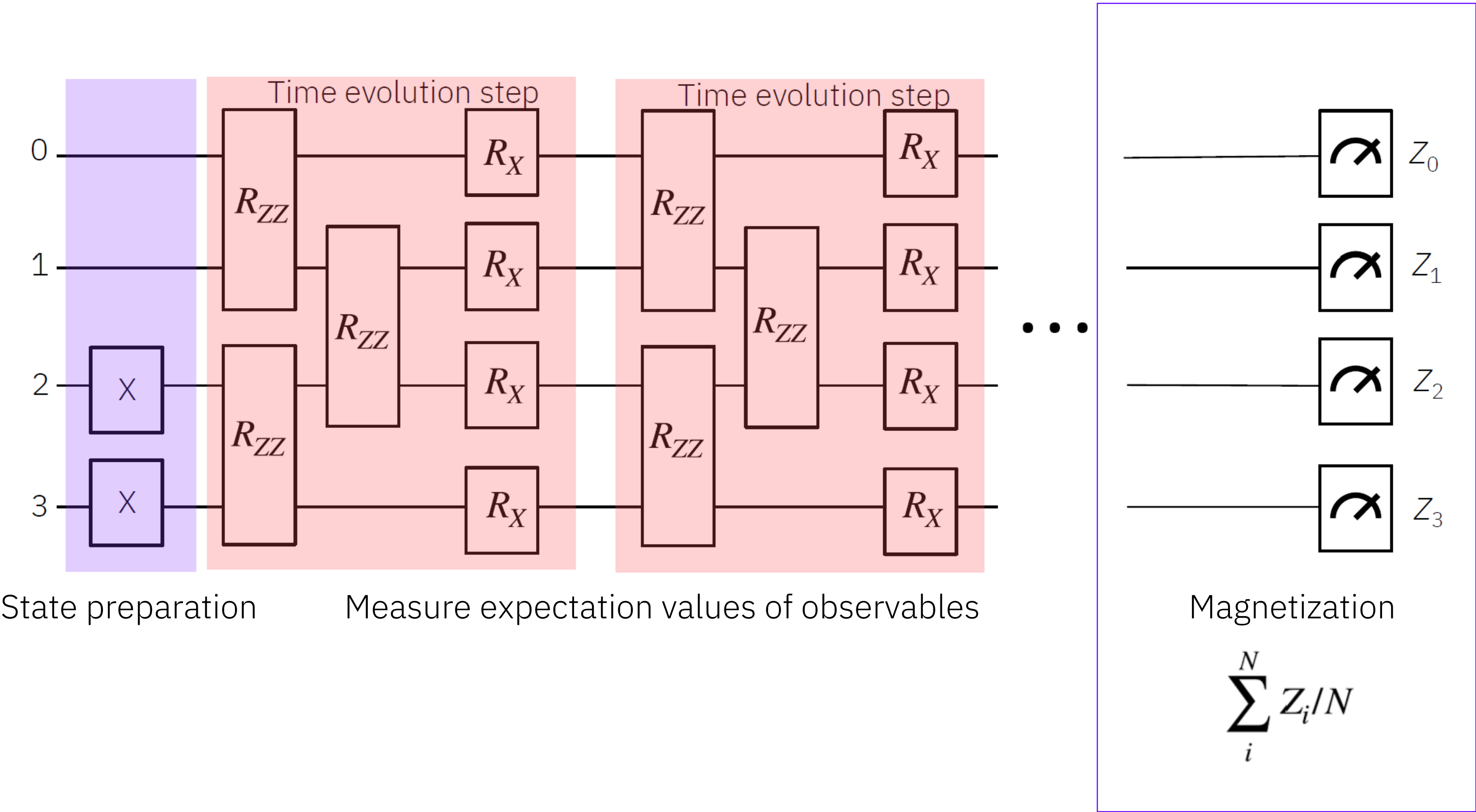
State preparation

By repeating this time step we can get the wavefunction of time t

$$|\Psi(t)\rangle = e^{-i\hat{H}t} |\Psi(0)\rangle$$

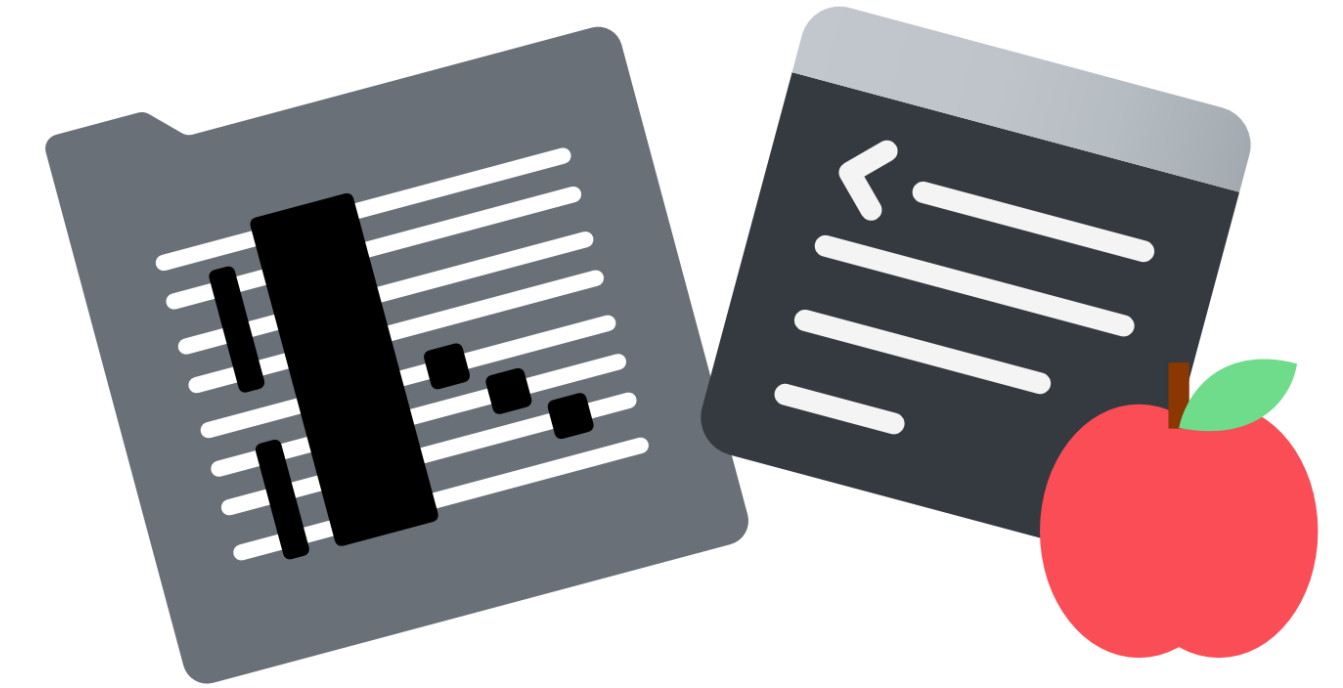
Practical Example: Trotterization (first-order)

Time evolution of 1D transverse-field Ising model



Y. Kim et al. *Evidence for the utility of quantum computing before fault tolerance.* Nature 618.7965 (2023)

Now let's code this up!



Further reading

Documentation:

<https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.library.PauliEvolutionGate>

Courses:

Quantum Simulation:

<https://quantum.cloud.ibm.com/learning/en/courses/utility-scale-quantum-computing/quantum-simulation>

Tutorials:

Multi-product formula (MPF) addon:

<https://quantum.cloud.ibm.com/docs/en/tutorials/multi-product-formula>